

Khang Tran

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Education

Princeton University, BA in Physics; minors in Computer Science and Engineering – Princeton, NJ Sept 2024 – May 2028

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Computer Graphics, Neural Rendering, Machine Learning & Pattern Recognition, Algorithms & Data Structures, Honors Linear Algebra, Multivariable Calculus

NUS High School of Mathematics and Science, Diploma (High Distinction) with Honours in Jan 2020 – Oct 2023
Engineering & Computer Science, and Majors in Mathematics, Chemistry & Physics – Singapore

- **GPA:** 4.9/5.0 — Computer Organization, Operating Systems (NUS dual enrollment)

Experience

Undergraduate Researcher, Roberto Car Group, Princeton University – Princeton, NJ June 2026 – present

- Built an auto-relaunching pipeline to run large-scale physics (DFT) simulations across multi-node HPC (SLURM) clusters, automatically recovering from job timeouts and failures.
- Automated job orchestration and parallelization across nodes to scale throughput.

Space Systems Engineer Intern, Aliena – Singapore Oct 2023 – Mar 2024

- Built an end-to-end Python software pipeline for real-time control, data acquisition, and visualization of laboratory instrumentation.
- Designed and built a hardware-in-the-loop tester – a programmable board emulating a Hall thruster’s signals across nominal and fault modes – to exercise the control-unit (ECU) software without live hardware in the loop.

Projects

Real-Time 3D Graphics Engine (C++ / OpenGL) July 2026 – present

- Built a real-time 3D rendering engine from scratch in **C++17 / OpenGL 3.3** — hand-written renderer, real-time shadow mapping, HDR/bloom pipeline, GPU-instanced particle system, impulse-based rigid-body solver with Coulomb friction, procedural terrain — all first-party, with **no engine, physics, or renderer library**. Cross-compiled to Windows via MinGW.
- Instrumented per-pass GPU timing (GL_TIME_ELAPSED queries on a 4-deep ring) behind a fixed-timestep, scripted-camera harness — **5.31 ms/frame at 1080p** (188 FPS mean, 137 FPS 1% low) on integrated graphics, with the 2048² shadow pass isolated at **0.469 ms**.
- Identified the renderer as **fill-bound rather than CPU- or draw-call-bound** with a resolution sweep — GPU time scaled 1 : 1.96 : 3.45 across 720p/1080p/1440p against a 1 : 2.25 : 4.0 pixel ratio, while CPU update stayed flat at ~0.13 ms.
- Benchmarked GPU instancing against an identical naive draw path — **4,149× fewer draw calls, 3.9× faster frames** at 50K particles, and **1M instanced grass blades (9.1M triangles) in one draw call**.

2D-to-3D Style Transfer for Reusable Assets Feb 2026 – May 2026

COS526: Neural Rendering (Princeton, Prof. Felix Heide)

- Developed a pipeline that turns a 3D mesh and a single style image into a fully stylized, game-ready asset with physically based rendering (PBR) textures (albedo, normal, roughness) baked via differentiable rendering.
- Built the multiview-consistent style-transfer stage — stylized 16 Kaolin-rendered views with SDXL + ConsisLoRA and depth/Canny ControlNets, enforcing cross-view consistency via cumulative view-warping, histogram matching, BiRefNet masking, and a RePaint inpainting schedule.

Skills

Programming: C/C++, Python, GLSL/WebGL, Git, MATLAB, Mathematica, LaTeX; data structures & algorithms (COS226)
Graphics & ML: Real-time rendering, ray tracing, shaders, image/signal processing (COS426); neural rendering & novel-view synthesis (COS526); PyTorch, GPU computing

Awards: **Pyka Prize for Physics Excellence**, Princeton Physics Dept. (2026); IBM Qiskit Hackathon Champion (Hardware track); **COSCON 2025 1st** (Undergraduate); Singapore Physics Olympiad Gold (2022)

Leadership & Teaching: CS Lab TA, COS217/226 (2026) & Physics Tutor, Princeton Physics Dept. (2025–26); Problem Setter & Organizer, Princeton Science Olympiad (2025–26); Problem Setter, Singapore Physics League / SPhL (2023–25)